

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 679 Data Visualization — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Munzner — Visualization Analysis and Design	The primary text. The what-why-how framework and the channel ranking this course uses come from here, and it is the most rigorous treatment available. Library copy; the slides are free from the author.
Ware — Information Visualization: Perception for Design, 4th edition	Modules 02 through 04. The perceptual research underlying every claim about which channels work, with the studies cited. Library copy.
Wilke — Fundamentals of Data Visualization (free online)	Modules 05, 09, and 10, free in full. Practical, opinionated, and unusually good on uncertainty and on the ways a chart misleads.
Tufte — The Visual Display of Quantitative Information	Module 10's conventions, and the data-ink argument. Influential, assertive, and worth reading critically — several of its claims did not survive measurement, which Module 05 §4 covers. Library copy.
Healey & Enns — Attention and Visual Memory in Visualization (free)	Module 02's preattentive material, with interactive demonstrations that are worth running rather than reading.
The IEEE VIS proceedings (free preprints)	The field's primary literature. Module 13's exercises use it; the evaluation sections are where the honest claims are.

Module-by-module map

Module 01 — Visualisation as Encoding

Resource	What it covers for this module
Munzner — Visualization Analysis and Design, chapters 1 through 3 (slides free)	§1 and §2 — the what-why-how framework and the mark-and-channel vocabulary, from the source.
Wilke — Fundamentals of Data Visualization, chapter 1 (free)	§1's mapping framing , with the aesthetics-as-channels vocabulary used throughout the book.
Anscombe — Graphs in Statistical Analysis (free)	§3's callout in the original — two pages, and it makes the argument once and permanently.
Matejka & Fitzmaurice — Same Stats, Different Graphs (free)	§3 extended — arbitrary shapes with identical statistics, generated, which is more striking than the original.

Module 02 — The Channel Ranking

Resource	What it covers for this module
Cleveland & McGill — Graphical Perception (free)	§1's ranking in the original — the experiments that established it, and still the reference.
Munzner, chapter 5 (slides free)	§1 through §3 — the ranking, separability, and the popout material, organised.
Heer & Bostock — Crowdsourcing Graphical Perception (free)	§1 replicated at scale — which confirmed the original ranking and extended it.
Ware — Information Visualization, chapters 1 through 5	§1 and §3's perceptual basis — why the ranking has the shape it does. Library copy.

Module 03 — Data Types and Marks

Resource	What it covers for this module
Munzner, chapters 2 and 3 (slides free)	§1 and §2 — the attribute types, the key/value distinction, and the dataset taxonomy.
Stevens — On the theory of scales of measurement (free)	§1's four types in the original — short, and the permitted-operations framing comes from here.
Wilkinson — The Grammar of Graphics	§2 and §3 formalised — the grammar that ggplot2 and Vega-Lite implement. Library copy.
Wilke, the chapters on directory of visualizations (free)	§2's counting rules applied — organised by what the data looks like rather than by chart name.

Module 04 — Colour

Resource	What it covers for this module
Wilke, the colour chapters (free)	The whole module — the scale types, the pitfalls, and the examples, free in full.
Borland & Taylor — Rainbow Color Map (Still) Considered Harmful (free)	§2's four defects , with the figures — short, and the false-boundary demonstration is the persuasive part.
Colorbrewer, and the viridis documentation (free)	§1 and §3 as working tools — with the colour-blind-safe and print-safe filters, which do §4's work for you.
Crameri, Shephard & Heron — The misuse of colour in science (free)	§2 measured across a literature — how prevalent the problem is, and what it has cost.

Module 05 — Chart Types

Resource	What it covers for this module
Wilke, the directory of visualisations and the pitfalls chapters (free)	§2 through §4 — the distribution charts and the axis conventions, with examples of each failure.
Tufte — The Visual Display of Quantitative Information	§3's data-ink argument in the original — read it alongside the replication work below. Library copy.
Bateman et al. — Useful Junk? (free)	§3's callout, measured — the study that found decoration aided recall without harming accuracy.
Cleveland — The Elements of Graphing Data	§4's banking argument and the aspect ratio material — where it was developed. Library copy.

Module 06 — Multiple Views and Interaction

Resource	What it covers for this module
Munzner, chapters 12 through 14 (slides free)	§1 and §2 — faceting, superimposing, and the view coordination design space.
Shneiderman — The Eyes Have It (free)	§2's task taxonomy and the overview-first sequence , in the original.
Robertson et al. — Effectiveness of Animation in Trend Visualization (free)	§3's callout, measured — small multiples against animation for comparison, with the numbers.
Heer & Shneiderman — Interactive Dynamics for Visual Analysis (free)	§2's vocabulary , organised as a taxonomy of interaction operations.

Module 07 — Networks and Hierarchies

Resource	What it covers for this module
Munzner, chapter 9 (slides free)	§1 through §3 — node-link, matrix, and the hierarchy layouts, with the trades stated.

Resource	What it covers for this module
Ghoniem, Fekete & Castagliola — Node-link vs matrix (free)	§2's trade, measured — which tasks each representation wins, with the crossover point.
Behrisch et al. — Matrix reordering methods (free)	§2's ordering, surveyed — the methods, since the ordering is the whole design.
Shneiderman — Tree-maps (free)	§3's treemap in the original , with the space-filling motivation that justifies the area encoding.

Module 08 — Maps

Resource	What it covers for this module
Munzner, chapter 8 (slides free)	§2's alternatives — the spatial representation design space, with the trades.
Brewer — Designing Better Maps	§1 and §3 — projections, classification, and colour for maps, from a cartographer. Library copy.
Openshaw — The Modifiable Areal Unit Problem	§3's second trap — the original statement, and it is a more serious problem than its obscurity suggests. Library copy.
Wilke, the geospatial chapters (free)	§1 and §2 practically , with the projection comparisons drawn out.

Module 09 — Uncertainty

Resource	What it covers for this module
Wilke, the uncertainty chapters (free)	The whole module , free — and the best available practical treatment, with every representation drawn.
Hullman et al. — Hypothetical Outcome Plots (free)	§2's animated representation, measured — and it reads better than static intervals, which is the finding.
Kay et al. — When (ish) is My Bus? (free)	§3's quantile dotplots in the original — the frequency framing applied to a real decision.
Belia et al. — Researchers misunderstand confidence intervals (free)	§2's misreading, measured on researchers — which is the relevant population and the sobering result.

Module 10 — Deception

Resource	What it covers for this module
Wilke, the pitfalls and the honesty chapters (free)	§1 and §2 , with the examples drawn and the axis arguments made from the channels.
Pandey et al. — How Deceptive are Deceptive Visualizations? (free)	§1 measured — how much a truncated axis actually changes a reader's judgement, which is a large amount.

Resource	What it covers for this module
Cairo — How Charts Lie	§2's selective framing in depth — and the best available treatment of the omission techniques. Library copy.
Huff — How to Lie with Statistics	§2 from 1954 , and still accurate — which is itself informative about how little the techniques have changed.

Module 11 — Scale

Resource	What it covers for this module
The datashader documentation and its principles (free)	§2 as a working system — and its discussion of count scaling is the best treatment of that specific problem.
Carr et al. — Scatterplot matrix techniques for large N (free)	§1 and §2 in the original — where hexagonal binning was introduced, for exactly this reason.
Liu, Jiang & Heer — imMens: Real-time Visual Querying of Big Data (free)	§4's data cube approach — precomputed aggregation for interactive latency.
Lins, Klosowski & Scheidegger — Nanocubes (free)	§4's structure — the aggregation index that makes the interaction a lookup.

Module 12 — Scientific Visualisation

Resource	What it covers for this module
Engel et al. — Real-Time Volume Graphics	§3 in full — the integral, the discretisation, and the GPU implementation. Library copy.
Levoy — Display of Surfaces from Volume Data (free)	§3 in the original — direct volume rendering as it was introduced, and the transfer function argument is already there.
Lorenson & Cline — Marching Cubes (free)	§2's isosurface extraction — and the connection to CSCE 620's geometry.
Cabral & Leedom — Line Integral Convolution (free)	§4's third technique in the original , with the construction explained.

Module 13 — Claiming a Visualisation Shows Something

Resource	What it covers for this module
Munzner, the validation chapter (slides free)	§1's four levels — and what kind of evidence each level requires, which is the rigorous version of this module.
Hullman & Gelman — Designing for Interactive Exploratory Data Analysis (free)	§1's reading problem — and the argument that a figure encodes a model whether or not you said so.
Wilke, the telling-a-story chapters (free)	§2's captions and annotation — with the put-it-in-the-figure argument made concretely.

Resource	What it covers for this module
The IEEE VIS evaluation literature (free preprints)	§4's step five, done properly — how the field evaluates figures, and the limitations sections are the model.